

DIPAK KUMAR

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Education

University of Ottawa

Ottawa, ON

Bachelor of Science in Computer Science (Honours)

Expected Graduation: April 2027

- Research Assistant, Mer Lab; Dance Lead, South Asian Club Ottawa; active member, Boxing Club.
- *Relevant Coursework:* Data Structures, Algorithms, Object-Oriented Programming, Databases, Operating Systems, Software Construction, Computer Networking, Statistics

Technical Skills

Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL, Bash

Development & Testing: Git, GitHub Actions, Jira, Jenkins, Docker, Pytest, Postman, CI/CD

Frameworks & Data: React, Flutter, FastAPI, Firebase, PostgreSQL, MS SQL, Pandas, NumPy

Cloud & AI: AWS, Azure, GCP, Kubernetes, Terraform, GitHub Copilot, Claude, LLMs

Experience

Deltek

May 2026 – Present

Software Engineering Intern (Co-op)

Ottawa, ON

- Developed **Python** automation and debugged **C/C++** application code to build regression and edge-case tests validating feature behavior, error handling, and software stability.
- Diagnosed software defects by reproducing failed scenarios, analyzing logs, tracing failures through application code, and partnering with developers to validate fixes before closing issues in **Jira**.
- Expanded automated regression coverage by **80%** by adding new test scenarios, reviewing **Git** changes, and using AI-assisted tools to identify edge cases and accelerate debugging within an Agile workflow.

LoyaltyValet

May 2025 – Aug 2025

Founding Engineer

Ottawa, ON

- Built and shipped full-stack features across **React**, **Flutter**, and **Firebase**, supporting a platform with **12,000+** registered users and roughly **\$120K ARR** across 8 retail clients.
- Developed a **React** administrative dashboard and automated merchant onboarding workflows, reducing setup time by **75%** and improving operational consistency across client accounts.
- Worked within a fast-moving startup environment to scope, debug, and iterate on production features, translating changing product requirements into reliable customer-facing functionality.

Mer Lab, University of Ottawa

Jan 2026 – Apr 2026

Research Assistant

Ottawa, ON

- Designed a containerized **Python** and **Docker** data pipeline for reproducible preprocessing and model training across **11 biological datasets**.
- Built automated data-validation workflows to detect inconsistencies, standardize preprocessing outputs, and improve reproducibility for research supporting a Faculty of Medicine publication.

Projects

Grid Runner | *Python, FastAPI, WebSockets, JavaScript* | **GitHub**

- Built a real-time **100-robot** warehouse simulator that keeps robots from colliding in shared aisles, even when a path gets blocked, and streams the floor live over **WebSockets**.

RAG Document Q&A Agent | *Python, FastAPI, FAISS, LLMs* | **GitHub**

- Built a modular **FastAPI** backend for document ingestion, vector indexing, semantic search, and cited LLM-based Q&A using **FAISS**, returning relevant responses in under **2 seconds**.

Interrupt-Driven Real-Time Alarm System | *C, Assembly, Dragon-12*

- Built and debugged an interrupt-driven embedded system in **C** and **Assembly**, integrating keypad, LCD, seven-segment display, speaker peripherals, and hardware timers for reliable real-time control.